

BATCH – 16

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Project Title :

Reverse Engineering Based Intelligent Cyber Threat Detection and Analysis Platform

Abstract :

Cybersecurity threats such as malware, phishing attacks, malicious URLs, suspicious applications, and unauthorized activities continue to pose significant challenges to organizations. The increasing volume and complexity of security events make it difficult for security teams to identify, analyze, and respond to threats effectively. Conventional security solutions often operate independently and focus on specific categories of threats, resulting in fragmented security information and limited visibility into the overall security posture of an organization. To address these challenges, this project proposes an Enterprise Cybersecurity Intelligence Platform that provides a centralized environment for collecting, analyzing, correlating, and visualizing cybersecurity threats. The proposed platform analyzes multiple sources of security data, including suspicious Android APK files, URLs, emails, and system or security logs. The APK analysis module utilizes reverse engineering and static malware analysis techniques to identify dangerous permissions, suspicious API calls, embedded URLs, hardcoded information, obfuscated code, and other Indicators of Compromise (IOCs). The URL and phishing detection modules analyze suspicious links, domains, redirects, impersonation attempts, and potentially malicious content.

The platform also integrates a Threat Intelligence and IOC Management mechanism to maintain information about known malicious domains, IP addresses, malware signatures, and attack patterns. Findings generated from different security modules are combined using a Cross-Source Threat Correlation and Risk Assessment Engine, which identifies relationships between security events and evaluates the overall severity of potential threats. Risk levels are determined based on factors such as threat severity, confidence, frequency, and potential organizational impact. A centralized cybersecurity intelligence dashboard provides visualization of detected threats, risk scores, security alerts, and incident information, enabling administrators to monitor the organization's security status from a single interface. Alert management and incident prioritization mechanisms help highlight critical threats and support timely response. Overall, the proposed Enterprise Cybersecurity Intelligence Platform aims to provide a unified, scalable, and effective approach for cyber threat analysis, threat intelligence correlation, risk assessment, and continuous security monitoring, thereby improving organizational cybersecurity awareness and incident response capabilities.

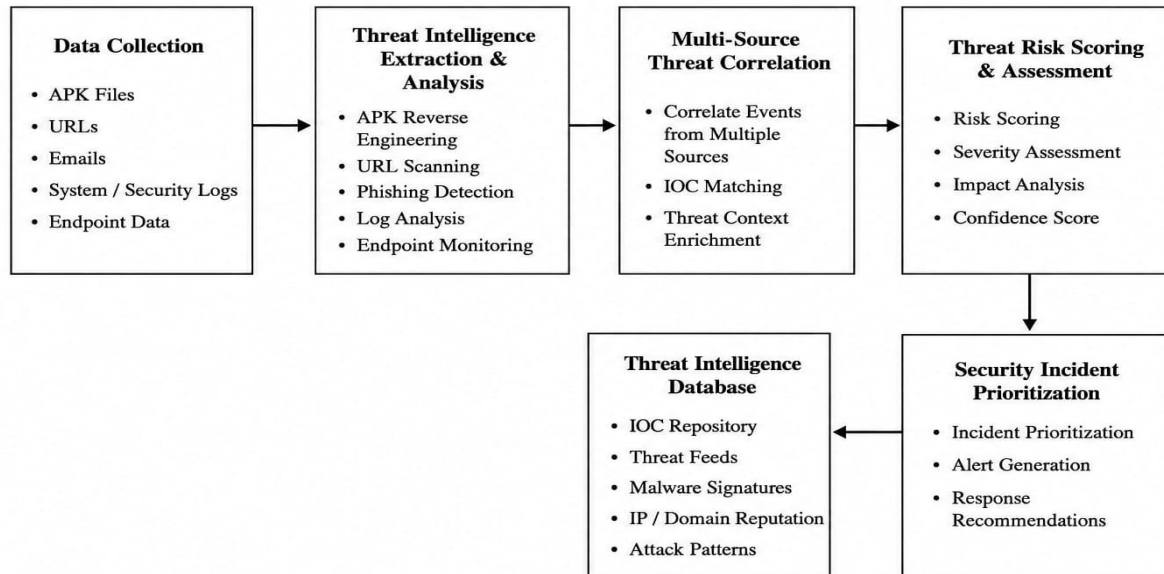
Keywords :

Enterprise Cybersecurity, APK Reverse Engineering, Static Malware Analysis, Cyber Threat Intelligence, Indicators of Compromise (IOC), Malicious URL Detection, Phishing Detection, Threat Correlation, Risk Assessment, Security Monitoring, Threat Intelligence Database, Cybersecurity Dashboard

Project Modules :

1. Security Data Acquisition and Preprocessing Module (SDAP)
2. APK Reverse Engineering and Malware Analysis Module (ARMA)
3. URL and Phishing Detection Module (UPD)
4. Threat Intelligence and IOC Management Module (TIIM)
5. Cross-Source Threat Correlation and Risk Assessment Module (CTRA)
6. Security Monitoring and Incident Prioritization Module (SMIP)
7. Cybersecurity Intelligence Visualization and Alert Management Module (CIVA)

Architecture Diagram :



Literature Review :

	Article	Year	Methodology	Research Gap
1	Malware Detection: A Framework for Reverse Engineered Android Applications	2022	Reverse Engineering, Static APK Analysis	Relies on manifest and permission-based detection only; lacks integration with cyber threat intelligence or risk correlation.
2	Android Malware Detection Using Static Analysis Techniques	2024	Static Analysis, Permission Analysis	Limited to static analysis; no URL/phishing detection or centralized security dashboard.
3	Malicious URL Detection Using Machine Learning	2024	URL Feature Analysis, Machine Learning	Focuses only on URL detection; not integrated with malware analysis or threat intelligence modules.
4	Phishing Website Detection Using URL and Content Analysis	2023	URL Similarity Analysis, HTML Content Analysis	Addresses phishing websites only; does not cover APK or system log analysis.
5	Cyber Threat Intelligence: Challenges and Opportunities	2024	Threat Intelligence Framework	Conceptual framework only; lacks implementation of reverse engineering or automated risk scoring.
6	Security Information and Event Management (SIEM): A Survey	2022	Event Correlation, Log Analysis	Survey-level analysis; no malware- or URL-specific detection modules.
7	Endpoint Detection and Response: Current Trends	2023	Endpoint Monitoring, Behavioral Analysis	Limited to endpoint devices; no APK reverse engineering or URL analysis capability.
8	Static Analysis Techniques for Android Malware Detection	2021	Static Code Analysis	Faster detection but limited to code inspection; lacks threat intelligence correlation.
9	Threat Intelligence Platform for Advanced Cybersecurity	2023	Threat Intelligence Database	Database-centric approach; lacks reverse engineering-based malware analysis.
10	Android Reverse Engineering for Malware Investigation	2023	APKTool, JADX, Reverse Engineering	Focused on investigation only; no correlation engine or risk visualization dashboard.